

The empirical recurrence for OEIS A280067, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A280067 records “Empirical recurrence of order 87” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 650$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{87} = 81 \neq 0$, so no further tail satisfies anything shorter; any order-87 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A280067 is “Number of $n \times 6$ 0..1 arrays with no element unequal to a strict majority of its horizontal, vertical and antidiagonal neighbors and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

5, 19, 104, 603, 3729, 23564, 145485, 915505, ...

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Dec 25 2016 06:40:15, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 650$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 650$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 87: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 87.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1300$ exact terms from $n = 5$ on, it returns order 87 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	12	c_{23}	−1530513799	c_{45}	−67125224560	c_{67}	−101435932178
c_2	−8	c_{24}	5107432746	c_{46}	−795825940797	c_{68}	−27131434998
c_3	−367	c_{25}	2033204267	c_{47}	1946950196972	c_{69}	34290520036
c_4	916	c_{26}	−7645995467	c_{48}	−815759262658	c_{70}	8089917747
c_5	4637	c_{27}	3697214014	c_{49}	−621991014242	c_{71}	−8046521339
c_6	−16338	c_{28}	−817665786	c_{50}	1609015538705	c_{72}	−1417695148
c_7	−34191	c_{29}	−8439073520	c_{51}	−1168510510243	c_{73}	310929664
c_8	165037	c_{30}	26375610512	c_{52}	266374069664	c_{74}	426110380
c_9	160339	c_{31}	−15263130111	c_{53}	−1772235428527	c_{75}	253476624
c_{10}	−1207951	c_{32}	−50490509356	c_{54}	903462816262	c_{76}	−137323843
c_{11}	−351908	c_{33}	92971322356	c_{55}	1645268911704	c_{77}	−50940237
c_{12}	6725509	c_{34}	13662959267	c_{56}	−1440043709145	c_{78}	20957706
c_{13}	−433651	c_{35}	−130188337197	c_{57}	244980458426	c_{79}	7192594
c_{14}	−30290497	c_{36}	−37750788504	c_{58}	840137161949	c_{80}	−1505759
c_{15}	7204690	c_{37}	210415306230	c_{59}	−101231680972	c_{81}	−315441
c_{16}	110665045	c_{38}	8744973261	c_{60}	−1173220452731	c_{82}	−149181
c_{17}	−29535148	c_{39}	−384799535450	c_{61}	306953105914	c_{83}	23322
c_{18}	−323210517	c_{40}	548856948927	c_{62}	518120194069	c_{84}	16101
c_{19}	−6344203	c_{41}	−45158665188	c_{63}	−262596917331	c_{85}	−1620
c_{20}	888428367	c_{42}	−374578185160	c_{64}	−149567277901	c_{86}	−460
c_{21}	422570624	c_{43}	47763244820	c_{65}	218977930185	c_{87}	81
c_{22}	−2357216406	c_{44}	12318511858	c_{66}	49818286288		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A280067 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{87} - c_1x^{87-1} - \dots - c_{87}$. Its constant term is $-c_{87} = -81$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 87 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 87, so $\deg q = 87$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 87, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 87, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 81.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-87 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A280067.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.